VISHNU DHARSAN A S

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SUMMARY

Software Engineer with 1+ year of experience building AI-powered applications using Python, FastAPI, LangChain, Ollama, PostgreSQL, MongoDB, and Retrieval-Augmented Generation (RAG). Experienced in designing scalable backend APIs, semantic search pipelines, prompt engineering, and AI assistants for industrial automation. Seeking Software Engineer, AI Engineer, or GenAI Engineer roles.

TECHNICAL SKILLS

Programming Languages: Python, C, SQL, Java, HTML, CSS, JavaScript

Frameworks & Libraries: LangChain, TensorFlow, Keras, Scikit-learn, NumPy, Pandas

LLM & AI Tools: Ollama, Hugging Face Transformers, OpenAI API, FAISS, FlaskAPI, FastAPI , Embeddings, Vector Databases, MCP, Semantic Search

Databases & Cloud: ChromaDB, PostgreSQL, MongoDB, MySQL,

Data Analytics & Visualization: Power BI, MS Excel, Matplotlib, Seaborn, Plotly

Software & DevOps Tools: Podman, Docker, Git, GitHub, Google Cloud Platform (GCP), VS Code, Jupyter

Concepts: Retrieval-Augmented Generation (RAG), Prompt Engineering, Context Management, Document Ingestion, AI Agents, System Design, Data Structures & Algorithms, Object-Oriented Programming (OOP)

EXPERIENCE

ASM Technologies (Software Engineer) *Aug 2025 – Present*

- Designed and developed an enterprise AI assistant for industrial failure analysis by building an end-to-end Retrieval-Augmented Generation (RAG) solution for troubleshooting, log analysis, and FA report generation.
- Built scalable document ingestion and semantic retrieval pipelines using Ollama LLMs, ChromaDB, and LangChain to enable accurate, context-aware responses from technical documentation and domain knowledge.
- Developed REST APIs using FastAPI and Flask, integrating PostgreSQL and MongoDB for efficient data management, chat history, and application workflows.
- Implemented prompt engineering, retrieval optimization, and context management to enhance response quality and reduce hallucinations, while enabling log analysis, root cause identification, and citation-backed failure analysis report generation for industrial troubleshooting.

Project Intern with AES technologies Oct 2023 – May 2024

HRIQ- HRMS SYSTEM

- Full stack Development of HRMS project for employee management with work on the Reimbursement feature.
- Analyzed employee data to identify trends and improve reimbursement processing efficiency using SQL and .NET.
- Collaborated with a cross-functional team of 6, utilizing jQuery, C#, and SQL Server to deliver a scalable and modular HR platform, decreasing feature deployment time by 20%.

PROJECTS

Automotive failure detection and Recovery Recommendation:

- Created a hybrid LLM + rules-based system that classified automotive failures with ~90% accuracy across 4 OEM datasets and generates recovery recommendations and created web UI.
- Compared outputs of different LLMs for optimal performance and selected the best-suited model.
- Python, LLMs – GPT-3, Llama 3, Bard, Claude, Falcon; static rule-based classification, React, NodeJS.

Bearing fault classification (Paper-published):

- Developed a deep learning model to predict the faults and lifetime of bearings used in machines.
- Python, models - CNN, ANN, TensorFlow, SVM, LightGBM

Full Stack E-Commerce App using Next.js:

- Developed a dynamic e-commerce platform using **Next.js**, **MongoDB**, and **Tailwind CSS**, enabling seamless product listing, cart, and checkout workflows.
- Created **RESTful APIs** for user sessions, product listings, and order management.

EDUCATION

CS academy Erode, TN
Class X - 92 % *May 2019*
Class XII - 94% *May 2021*

PSG College of Engineering Coimbatore, TN
Bachelor of Engineering - Computer Science and Engineering - ***8.0 CGPA*** *Nov 2021 – April 2025*

ADDITIONAL

- Represented ASM technologies in CII INNOVERGE 2025.
- Presented Technical Talk on ”Building Rag Chatbot using NVIDIA NIM”.
- Entrepreneur’s club - Secretary (Event organisation, Speaker sessions, Overall management)
- Computer Science and Engineering Association – Photography and publicity head
- Class representative -5th semester
- Sports – Badminton, Cricket and Swimming